e	$k=2$	$k=4$	$k=8$	Avg.
4	0.6187	0.6300	0.6566	0.6351
8	**0.6280**	**0.6414**	0.6721	**0.6472**
16	0.6195	0.6400	**0.6736**	0.6444
32	0.6039	0.6209	0.6477	0.6242
64	0.5597	0.5729	0.5843	0.5723
all	0.5723	0.5771	0.5828	0.5774

Table 2: Recall@1000 results on the test set with varying number of expansion terms e from each relevant document. Results are averaged over the shuffles.

it will rank documents higher that are strongly preferred by one ranking model than documents that are weakly preferred by multiple models.[9]

6 Results

6.1 2nd Stage Retrieval: Query Expansion

We report recall@1000 results of the second stage retrieval with varying number of expansion terms e in Table 2. Note that by increasing e the performance increases, reaching a maximum at $e = 8$. However, when further increasing e, the recall drops. We observe qualitatively that extracting more terms per document also includes more non-specific terms or even stop words which hurt performance. Based on the recall@1000 performance on the validation set (see Appendix I) we use the documents obtained by extracting $e = 16$ terms for the final re-ranking step. In this work, we do not focus on the first stage retrieval. For completeness, we report the results in Appendix B.

6.2 Re-Ranking Performance

We report the nDCG@20 ranking performance in Table 3 and additional zero-shot baselines in Appendix C. We first note that increasing the amount of relevance feedback k generally improves performance. Furthermore, we observe that BM25-QE already performs well. Neither the kNN approach, nor the Cross-Encoder zero-shot and Query FT, nor the wide variety of zero-shot models are able to outperform BM25-QE, except on TREC-Covid. We note a superior performance on Webis-Touché, although this task is the most challenging for neural models in our test suite. This agrees with related work that indicates that BM25 beats all other methods on this task (Thakur et al., 2021). When looking at the CE experiments, we observe incremental

[9]For example, if $h_0(d_0) = 5$, $h_1(d_0) = 15$, $h_0(d_1) = 10$ and $h_1(d_1) = 10$ then $s_0 > s_1$.

	Robust	Covid	News	Touché	Avg.
BM25-QE					
$k=2$	0.4480	0.5632	0.3846	0.2602	0.4140
$k=4$	0.4843	0.6079	0.3877	0.2558	0.4339
$k=8$	0.5568	0.6606	0.4049	**0.2982**	0.4801
Avg.	0.4964	0.6106	0.3924	0.2714	0.4427
kNN					
$k=2$	0.4259	0.6736	0.3492	0.1646	0.4033
$k=4$	0.4342	0.6789	0.3539	0.1697	0.4092
$k=8$	0.4698	0.7069	0.3925	0.1904	0.4399
Avg.	0.4433	0.6865	0.3652	0.1749	0.4175
CE Zero-Shot					
$k=2$	0.3937	0.6917	0.2955	0.1731	0.3885
$k=4$	0.4185	0.7018	0.3189	0.1767	0.4040
$k=8$	0.4335	0.7150	0.3285	0.1799	0.4142
Avg.	0.4152	0.7028	0.3143	0.1766	0.4022
CE Query-FT					
$k=2$	0.4375	0.6833	0.2942	0.1887	0.4009
$k=4$	0.4786	0.7182	0.3463	0.2080	0.4378
$k=8$	0.5376	**0.7677**	0.3645	0.1975	0.4668
Avg.	0.4846	0.7231	0.3350	0.1981	0.4352
CE MAML + Query FT					
$k=2$	0.4529	0.7129	0.2526	0.2212	0.4099
$k=4$	0.5079	**0.7498**	0.3358	0.2292	0.4557
$k=8$	0.5572	0.7449	0.3557	0.2201	0.4695
Avg.	0.5060	0.7359	0.3147	0.2235	0.4450
Rank Fusion: kNN & BM25-QE					
$k=2$	0.4635	0.6903	0.3783	0.2263	0.4396
$k=4$	0.5020	0.6858	**0.4228**	0.2438	0.4636
$k=8$	0.5574	0.7470	**0.4359**	0.2744	0.5037
Avg.	0.5076	0.7077	**0.4123**	0.2482	0.4689
Rank Fusion: CE MAML + Query FT & BM25-QE					
$k=2$	**0.5164**	**0.7269**	**0.3934**	**0.2670**	**0.4759**
$k=4$	**0.5576**	0.7449	0.4084	**0.2701**	**0.4953**
$k=8$	**0.6380**	0.7489	0.4148	0.2809	**0.5207**
Avg.	**0.5707**	**0.7402**	0.4055	**0.2727**	**0.4973**

Table 3: nDCG@20 test set results averaged over three seeds with a varying number of feedback documents (k). In bold, the best performing model, the runner-up is underlined.

performance increases when the relevance feedback is integrated. CE zero-shot is outperformed by query fine-tuning, which is subsequently outperformed when MAML training is added. This shows that our proposed direct integration of relevance feedback in the model is effective and that the parameters obtained by MAML training are better able to adapt to new queries given the relevance feedback. This method also slightly outperforms BM25-QE.

Finally, combining the rankings of the lexical retrieval and neural re-ranker is particularly effective. While different methods excel at each dataset (e.g. BM25 on Webis-Touché or neural models on